

MILESTONE 1

Platform Foundation & Authentication System

Enterprise Learning Platform – Skill & Career Guidance System

1. Milestone Overview

Milestone 1 focuses on building the **foundation of the Enterprise Learning Platform** and implementing a complete and secure **student authentication system**.

The milestone establishes the initial user interface, navigation structure, student registration and login functionality, database user management, JWT-based authentication, authorization middleware, and protected API access.

Milestone 1 Includes

- Landing/Home Page
 - Student Registration
 - Student Login
 - Google Authentication Interface
 - Student Role Selection
 - Form Validation
 - Password Security
 - Database User Management
 - JWT Authentication
 - JWT Verification Middleware
 - Protected API Routes
 - Role-Based Authorization Foundation
 - Logout and Token Management
 - Responsive UI/UX
 - Authentication Error Handling
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2. Project Objective

The objective of Milestone 1 is to create a secure and user-friendly entry point to the platform.

The platform will eventually provide students with:

- Career guidance
- Skill assessments
- Learning roadmaps
- Course modules
- Module badges
- Internships
- QR-based certificates
- Personalized career recommendations

Milestone 1 provides the **technical and authentication foundation** required to build these features in subsequent milestones.

3. Landing / Home Page

The landing page introduces the platform and communicates its purpose to students.

Navigation Bar

The navigation bar should contain:

- Platform Logo
- Home
- Features
- Roadmaps
- About
- Theme/Light Mode Control
- Login
- Get Started

Hero Section

Main Heading

Build Your Future with Real Internships

Supporting Text

Personalized career guidance, learning roadmaps, and skill assessments for students — all in one connected platform.

Primary Actions

- Get Started
- Login

Trust & Social Proof

The landing page should display platform credibility through:

- Student/community count
- Participating campus count
- Student rating
- Achievement/certificate indicators

Feature Preview

The home page should introduce the major platform features:

- Course Modules
 - Module Badges
 - QR Certificates
 - Career Roadmaps
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4. Student Registration

The registration module allows new students to create an account.

Registration Fields

The following fields are required:

- Full Name
- Email Address
- Password
- Confirm Password

Registration Validation

The system must validate:

- Full Name is not empty
- Email is valid
- Password is not empty
- Password meets security requirements
- Confirm Password matches Password
- Terms & Conditions are accepted

Registration Actions

- Register
 - Continue with Google
 - Back to Login
 - Back to Home
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5. Student Login

The login module allows registered students to access the platform.

Login Fields

Login As

Initial role:

Student

The system should be designed so additional roles can be introduced later.

Email

- Required
- Valid email format

Password

- Required
- Password masking
- Show/Hide password option

Additional Features

- Remember Me
 - Forgot Password
 - Login
 - Continue with Google
 - Register
 - Back to Home
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6. Form Validation

All authentication forms must provide appropriate validation.

Registration Validation

- Required field validation
- Email format validation
- Password strength validation
- Confirm-password matching
- Terms & Conditions validation

Login Validation

- Required email validation
- Email format validation
- Required password validation
- Invalid credential handling

Validation messages should be clear and understandable to students.

7. Database User Management

A user/student record must be created in the database during registration.

The user model should contain appropriate fields such as:

- User ID
- Full Name
- Email
- Password Hash
- Role
- Account Status
- Created Date
- Updated Date

Additional authentication-related fields can be added as required.

Password Security

Passwords must **never be stored in plain text**.

The backend must securely hash passwords before storing them in the database.

8. JWT Authentication

Milestone 1 includes **JWT (JSON Web Token) based authentication**.

JWT will be used to authenticate users when they access protected resources.

Login Authentication Flow

1. Student enters email and password.
 2. Frontend sends credentials to the backend.
 3. Backend validates the request.
 4. Backend finds the user in the database.
 5. Password is verified against the stored password hash.
 6. If credentials are valid, the backend generates a JWT.
 7. JWT is returned to the authenticated client.
 8. Client securely handles the authentication token.
 9. The token is included with requests to protected APIs.
 10. Backend validates the token before granting access.
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9. JWT Token

The JWT should contain only the information required for authentication and authorization.

Possible claims include:

- **sub** – User ID
- **role** – User role
- **iat** – Token issued time
- **exp** – Token expiration time

Important Security Rule

Passwords and other sensitive information must **never** be stored inside the JWT.

The JWT signing secret or private signing key must remain securely on the backend and must never be exposed to the frontend.

10. JWT Authentication Middleware

Protected APIs must use authentication middleware.

The middleware should:

1. Receive the API request.
2. Extract the JWT.
3. Verify that a token exists.
4. Validate the JWT signature.
5. Check token expiration.
6. Decode the required user information.
7. Identify the authenticated user.
8. Pass the authenticated request to the protected API.

If authentication fails, the API must reject the request.

11. Protected APIs

JWT authentication will protect APIs that require an authenticated student.

Examples include:

- Student Profile
- Student Dashboard
- Skill Assessments
- Course Modules
- Learning Progress
- Career Roadmaps
- Internships
- Module Badges
- Certificates
- Personalized Recommendations

These features can be implemented in later milestones, but Milestone 1 establishes the authentication mechanism required to protect them.

12. Role-Based Authorization

Milestone 1 establishes the foundation for role-based authorization.

Initial Role

Student

The system should be designed to support additional roles in future milestones, such as:

- Admin
- Mentor

Authorization middleware can use the authenticated user's role to determine whether access to a particular API or resource is permitted.

13. Unauthorized Access Handling

The backend must reject requests when:

- JWT is missing
- JWT is invalid
- JWT signature is incorrect
- JWT has expired
- User account is inactive/invalid
- User does not have the required role

The system should return appropriate authentication/authorization responses without exposing sensitive backend information.

14. Logout & Token Management

The application must provide a logout mechanism.

When a user logs out:

1. The authenticated client state is cleared.
2. The authentication token is removed from the client.
3. The user is redirected to the Login or Home page.

JWT expiration should prevent tokens from remaining valid indefinitely.

If refresh tokens are introduced, they should be securely managed separately from access tokens.

15. Google Authentication

Both Login and Registration screens should provide:

Continue with Google

The authentication flow should support:

1. User selects Google authentication.
2. User authenticates with Google.
3. Backend verifies the authentication response.
4. User account is created or retrieved.
5. The user receives an authenticated session/token.
6. User proceeds to the platform.

If Google authentication is not fully integrated during Milestone 1, the UI and backend architecture should be prepared for future integration.

16. Navigation Flow

The following navigation must be implemented:

Home → Login

Login button opens the Login page.

Home → Registration

Get Started opens the Registration page.

Login → Registration

Register opens the Registration page.

Registration → Login

Back to Login returns to Login.

Authentication → Home

Back to Home returns to the Landing page.

Login → Forgot Password

Forgot Password opens the password recovery flow.

17. UI / UX Requirements

The platform should maintain a consistent visual design across all screens.

Design Characteristics

- Modern and clean interface
- Green/mint visual theme
- Rounded cards
- Rounded input fields
- Soft shadows
- Clear typography
- Consistent spacing

- Accessible controls
- Student-friendly language
- Responsive layouts

Components

The following should maintain a consistent design:

- Logo
 - Navigation
 - Buttons
 - Input fields
 - Cards
 - Icons
 - Form labels
 - Error messages
 - Success messages
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18. Responsive Design

The application must work across:

- Desktop
- Laptop
- Tablet
- Mobile

The interface should prevent:

- Overlapping elements
 - Text cutoff
 - Broken forms
 - Horizontal scrolling
 - Inaccessible buttons
 - Incorrect responsive layouts
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19. Security Requirements

Security is a major component of Milestone 1.

The system must:

- Hash passwords before database storage
 - Never store plain-text passwords
 - Use secure JWT signing secrets/keys
 - Never expose JWT signing secrets to the frontend
 - Use token expiration
 - Validate authentication tokens
 - Protect private APIs
 - Validate user input
 - Handle authentication errors securely
 - Use HTTPS in production
 - Avoid exposing sensitive information through error messages
 - Securely manage authentication tokens on the client
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20. Complete Authentication Flow

The complete Milestone 1 authentication architecture is:

Student



Landing Page



Registration / Login



Frontend Validation



Authentication API



Backend Authentication Service



Database User Verification



Password Hash Verification



JWT Generation



JWT Returned to Client



Authenticated Student



Protected API Request



JWT Authentication Middleware



JWT Verification



User & Role Verification



Protected Platform Resource



Logout



Authentication State Cleared

21. Milestone 1 Deliverables

Frontend

- Landing/Home page
- Navigation bar
- Login page
- Registration page
- Student role selection
- Form validation
- Password visibility control
- Remember Me
- Forgot Password interface
- Google authentication interface
- Responsive layouts
- Authentication navigation

Backend

- Registration API
- Login API
- Password hashing
- JWT generation
- JWT verification
- JWT authentication middleware
- Protected API routes

- User identity extraction
- Role-based authorization foundation
- Authentication error handling
- Logout/token management

Database

- User/Student model
 - User ID
 - Full Name
 - Email
 - Secure Password Hash
 - User Role
 - Account Status
 - Created/Updated timestamps
 - Appropriate database constraints
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23. Milestone 1 Final Outcome

At the completion of Milestone 1, the Enterprise Learning Platform will have a **complete frontend and secure authentication foundation**.

The system will provide:

Landing Page + Registration + Login + Database + Password Security + JWT Authentication + Authorization Middleware + Protected APIs + Logout + Responsive UI

This foundation enables the development of the next platform modules.

Final Milestone Statement

Milestone 1 establishes the complete foundation of the Enterprise Learning Platform by combining the initial UI/UX implementation with secure student

registration, login, database management, JWT authentication, authorization, protected APIs, and responsive user experience.